EXPERIMENT 12: PERFORM PERSPECTIVE TRANSFORMATION ON THE IMAGE

Aim

To perform perspective transformation on an image using Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.

2. Read the input image.

3. Select four source points.

4. Select four destination points.

5. Compute perspective transformation matrix using cv2.getPerspectiveTransform().

6. Apply cv2.warpPerspective().

7. Display and save the transformed image.

Python Code

```
import cv2
import numpy as np

# Read input image
img = cv2.imread("input.jpg")

if img is None:
    print("Error: Image not found")
    exit()

rows, cols = img.shape[:2]

# Source points
pts1 = np.float32([
    [0, 0],
    [cols-1, 0],
    [0, rows-1],
    [cols-1, rows-1]
])
```

```
# Destination points
pts2 = np.float32([
    [50, 50],
    [cols-100, 20],
    [80, rows-50],
    [cols-30, rows-30]
])

# Perspective Transformation Matrix
matrix = cv2.getPerspectiveTransform(pts1, pts2)

# Apply Perspective Transformation
output = cv2.warpPerspective(img, matrix, (cols, rows))

# Save output
cv2.imwrite("perspective_output.jpg", output)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Perspective Transformation", output)

print("Perspective Transformation Matrix:")
print(matrix)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input

Output

Result

Thus, the perspective transformation was successfully performed on the input image using OpenCV.